

# Arnitel<sup>®</sup> ECO L460

## TPC

40% Renewable Content, Injection Molding, Food Contact Quality

Print Date: 2023-03-16

### Sustainability

Bio-based – 14C measurable

| Properties                             | Typical Data | Unit                   | Test Method       |
|----------------------------------------|--------------|------------------------|-------------------|
| <b>Rheological properties</b>          | <b>Value</b> |                        |                   |
| Melt volume-flow rate                  | 39           | cm <sup>3</sup> /10min | ISO 1133          |
| Temperature                            | 230          | °C                     | ISO 1133          |
| Load                                   | 2.16         | kg                     | ISO 1133          |
| Molding shrinkage (parallel)           | 1.3          | %                      | ISO 294-4         |
| Molding shrinkage (normal)             | 1.5          | %                      | ISO 294-4         |
| Molding shrinkage [parallel]           | 1.3          | %                      | Sim. to ISO 294-4 |
| Molding shrinkage [normal]             | 1.5          | %                      | Sim. to ISO 294-4 |
| <b>Mechanical properties</b>           | <b>Value</b> |                        |                   |
| Shore D Hardness (3s)                  | 45           | -                      | ISO 868           |
| Tensile modulus                        | 90           | MPa                    | ISO 527-1/-2      |
| Stress at break                        | 17           | MPa                    | ISO 527-1/-2      |
| Nominal strain at break                | 250          | %                      | ISO 527-1/-2      |
| Stress at 5% strain                    | 5.2          | MPa                    | ISO 527-1/-2      |
| Stress at 10% strain                   | 8.3          | MPa                    | ISO 527-1/-2      |
| Stress at 50% strain                   | 11           | MPa                    | ISO 527-1/-2      |
| Stress at 100% strain                  | 12           | MPa                    | ISO 527-1/-2      |
| Charpy notched impact strength (+23°C) | N            | kJ/m <sup>2</sup>      | ISO 179/1eA       |
| Izod notched impact strength (+23°C)   | N            | kJ/m <sup>2</sup>      | ISO 180/1A        |
| Izod notched impact strength (-30°C)   | 2.6          | kJ/m <sup>2</sup>      | ISO 180/1A        |

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| Properties                                     | Typical Data | Unit              | Test Method            |
|------------------------------------------------|--------------|-------------------|------------------------|
| Flexural modulus                               | 95           | MPa               | ISO 178                |
| Compression Set under constant strain at 70 °C | 45           | %                 | ISO 815                |
| <b>Mechanical properties (die cutting)</b>     |              | Value             |                        |
| Stress at break (normal)                       | 24           | MPa               | ISO 527-1/-2           |
| Tear strength (normal)                         | 107          | kN/m              | ISO 34-1; Method B     |
| Tear strength (parallel)                       | 106          | kN/m              | ISO 34-1; Method B     |
| Strain at break (normal)                       | 700          | %                 | ISO 527-1/-2           |
| <b>Thermal properties</b>                      |              | Value             |                        |
| Melting temperature (10°C/min)                 | 180          | °C                | ISO 11357-1/-3         |
| Vicat softening temperature (50°C/h 50N)       | 38           | °C                | ISO 306                |
| Burning Behav. at 0.75 mm nom. thickn.         | HB           | class             | IEC 60695-11-10        |
| Thickness tested                               | 0.75         | mm                | IEC 60695-11-10        |
| Burning Behav. at 3.0 mm nom. thickn.          | HB           | class             | IEC 60695-11-10        |
| Thickness tested                               | 3            | mm                | IEC 60695-11-10        |
| <b>Electrical properties</b>                   |              | Value             |                        |
| Volume resistivity                             | >1E13        | Ohm*m             | IEC 62631-3-1          |
| Surface resistivity                            | >1E15        | Ohm               | IEC 62631-3-2          |
| Electric strength                              | 20           | kV/mm             | IEC 60243-1            |
| <b>Other properties</b>                        |              | Value             |                        |
| Density                                        | 1140         | kg/m <sup>3</sup> | ISO 1183               |
| Humidity absorption                            | 0.03         | %                 | Sim. to ISO 62         |
| Biobased content (in polymer)                  | 43           | % (Bio C/Total C) | ASTM D6866-12 Method B |

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